Unit 8 Notes

Circles

How to use this file: Copy the notes carefully, then cover the worked examples and redo them without looking. For practice files, work first, then check the answer bank. Detailed solutions are included for Unit 3 through Unit 8 practice as requested.

Aligned to the Honors Geometry unit outline provided by the user.

Unit 8 Notes: Circles

Big goal

Circle geometry connects arcs, angles, chords, tangents, secants, and coordinate equations. The radius is the key: it defines the circle, creates right angles with tangents, and leads to the equation of a circle.

1. Circle vocabulary

A circle is the set of all points in a plane equidistant from a fixed point called the center. A radius connects the center to a point on the circle. A chord has both endpoints on the circle. A diameter is a chord through the center, and its length is $2r$. A secant intersects a circle at two points. A tangent intersects a circle at exactly one point.

2. All circles are similar

Any circle can be mapped to any other circle by a translation moving one center to the other and a dilation with scale factor equal to the ratio of the radii. Therefore all circles are similar.

3. Radii, chords, and tangents

- A radius perpendicular to a chord bisects the chord.
- The perpendicular from the center to a chord bisects the chord.
- Congruent chords are equidistant from the center; chords equidistant from the center are congruent.
- A radius drawn to a tangent point is perpendicular to the tangent line.
- Tangent segments from the same external point are congruent.

4. Arc and angle relationships

Angle type	Measure relationship
Central angle	Equal to its intercepted arc.
Inscribed angle	Half its intercepted arc.
Angle formed by tangent and chord	Half its intercepted arc.
Two chords intersect inside circle	Half the sum of intercepted arcs.
Two secants, two tangents, or secant and tangent outside circle	Half the difference of intercepted arcs.
Angle inscribed in a semicircle	Right angle.

5. Inscribed polygons

A polygon is inscribed in a circle if all vertices lie on the circle. A quadrilateral inscribed in a circle is cyclic, and opposite angles are supplementary. A polygon is circumscribed about a circle if each side is tangent to the circle.

6. Segment relationships

When chords, secants, and tangents create segments, products are equal.

Circle segment product formulas

Intersecting chords: (part)(part) = (part)(part).
Secant-secant from outside: (external)(whole) = (external)(whole).
Tangent-secant from outside: $(\text{tangent})^2 = (\text{external secant})(\text{whole secant})$.

7. Equation of a circle

A circle with center (h, k) and radius r has equation

$$(x - h)^2 + (y - k)^2 = r^2.$$

This comes from the distance formula: every point (x, y) on the circle is distance r from the center.

To graph a circle, plot the center, move r units up, down, left, and right, and sketch the circle through those points.

8. Point location

For center (h, k) and point (x, y), compare $(x - h)^2 + (y - k)^2$ to r^2:

- equal means the point is on the circle;
- less than r^2 means inside;
- greater than r^2 means outside.

9. Tangent line in the coordinate plane

A tangent line is perpendicular to the radius at the point of tangency. To write the tangent line:

1. Find the slope of the radius from center to tangent point.
2. Take the negative reciprocal for the tangent slope.
3. Use point-slope form through the tangent point.

10. Constructions

The circumcenter of a triangle is the intersection of the perpendicular bisectors of the sides; it is the center of the circumscribed circle. The incenter is the intersection of the angle bisectors; it is the center of the inscribed circle. A regular hexagon inscribed in a circle has side length equal to the radius.

11. Worked examples

Example 1: Equation

Center $(3, -2)$, radius 5:
$$(x-3)^2 + (y+2)^2 = 25.$$

Example 2: Tangent line

For center $(0, 0)$ and tangent point $(3, 4)$, the radius slope is $4/3$. The tangent slope is $-3/4$. The tangent line is
$$y - 4 = -\frac{3}{4}(x-3).$$

Example 3: Tangent-secant

A secant has external part 6 and internal part 18, so the whole secant is 24. The tangent length t satisfies
$$t^2 = 6(24) = 144,$$
so $t = 12$.

Mastery checklist

You are ready for Unit 8 when you can use arc-angle formulas, solve chord/secant/tangent segment equations, prove tangent-radius perpendicular facts, write and graph circle equations, classify points, and find tangent lines using slopes.